

TEN KINGS MATHEMATICAL GRADING - GEOMETRY TARGET V1

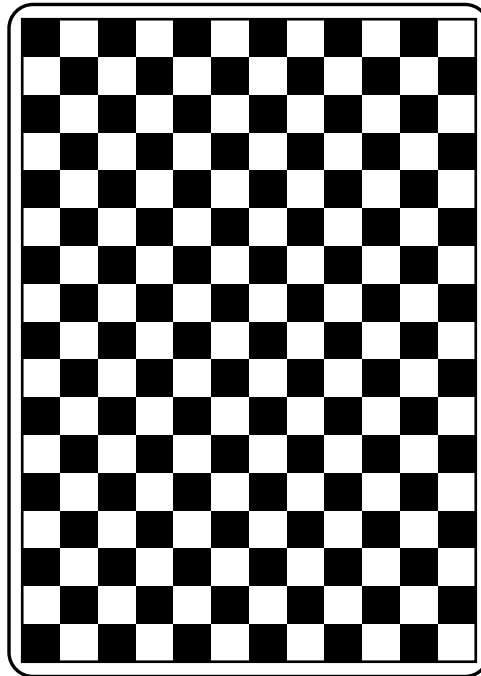
Print at Actual size / 100%.

Disable Fit, Shrink, and borderless expansion.

Accept only when each bar error plus measurement U95 is ≤ 0.20 mm.

Verify scale, then cut only the rounded 63.50 x 88.90 mm coupon outline.

After cutting, verify both coupon dimensions with recorded ruler U95.

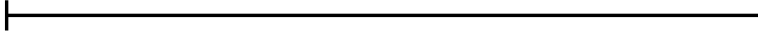


12 x 17 cells; 5.00 mm per cell; 11 x 16 internal corners.

50.00 mm X



100.00 mm X



200.00 mm Y



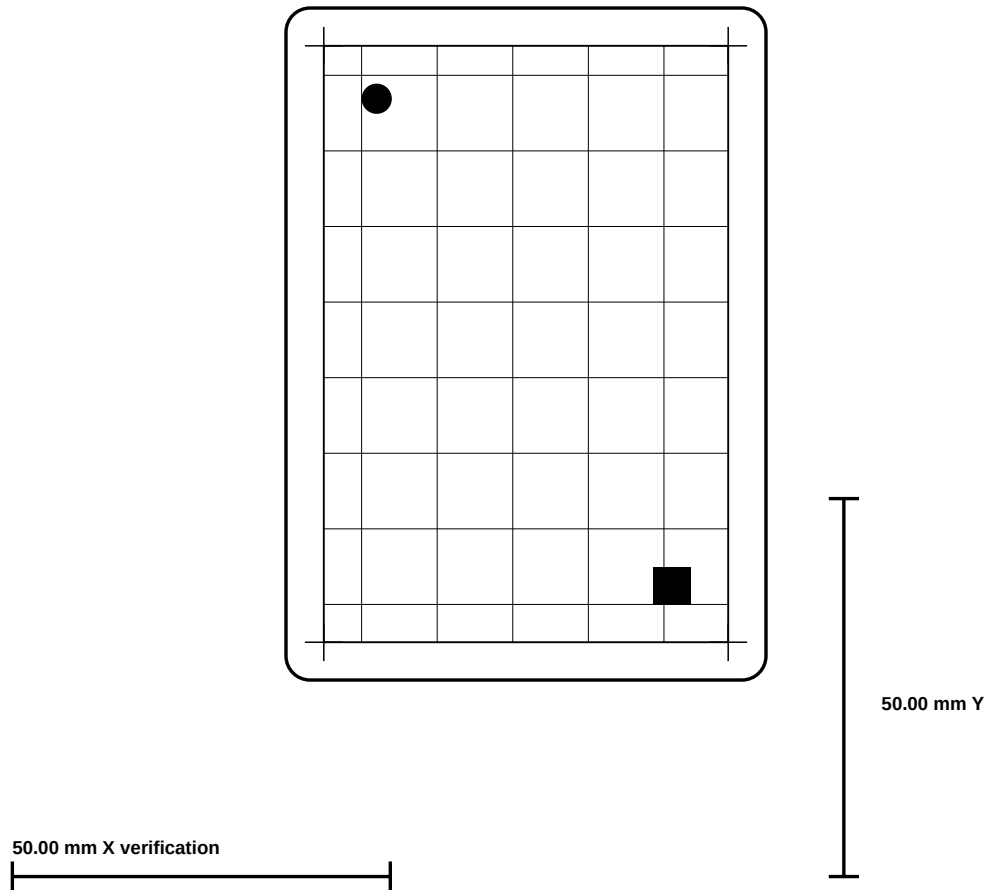
Capture at least 10 distinct in-frame poses at the fixed card plane.

Retain every source image, detected corner, distortion fit, and residual.

TEN KINGS MATHEMATICAL GRADING - CARD REFERENCE V1

Non-production calibration reference. Never grade or publish this target.

Outer cut: 63.50 x 88.90 mm. Printed frame inset: 5.00 mm per side.



Registration notes:

The circle and square intentionally break rotational symmetry.

Use the printed-frame contour for printed_border_v1 tests.

Hash this exact artifact before registered-design use.

TEN KINGS MATHEMATICAL GRADING - CALIBRATION RECORD V1

This target is metrology evidence, not proof by itself.

1. PRINT SCALE

- Print pages 1-2 at Actual size / 100%; do not fit or borderlessly expand.
- Measure the 100 mm X and 200 mm Y bars; error plus U95 must be ≤ 0.20 mm.
- Record measured axes, fixed-ruler ID/U95, operator, UTC time, and PDF SHA-256.

2. GEOMETRY

- Capture page 1 at the working plane across the required lens poses.
- Fit distortion and pixel/mm scale; retain every view and residual.
- Measure the cut coupon at 63.50 x 88.90 mm; error plus U95 must be ≤ 0.20 mm.
- Normalize page-1 checkerboard views; retain the fixed-transform holdout residual.
- Page 2 is only for centering/reference validation, never checkerboard normalization.

3. PHOTOMETRY

- Use the blank matte reverse only if neutral and free of print show-through; hash it.
- Capture at least 3 dark, flat-field, and pattern frames for channels 1-8.
- Measure each fixed light source point with the ruler at least 3 times.
- Retain geometry direction, irradiance validation, confidence, and exact hashes.

4. REPEATABILITY AND U95

- Remove and replace the target at least ten times without changing the rig.
- Repeat linear, area, relief, roughness, and neutral-patch Delta-E measurements.
- Compute certified U95; retain sample values and calculation method.

5. ACCEPTANCE

- Evaluate every centralized calibration-acceptance threshold.
- A failed or missing gate keeps status non-finalized and isCalibrated false.
- Never substitute a production card, internet image, or manual grade.

Operator: _____ UTC: _____

Printer/media: _____

Measured 100 X / 200 Y / U95: _____

Fixed ruler ID: _____

Artifact/profile hash: _____